

Erdős Problem #65

EP-65

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let G be a graph with n vertices and kn edges, and $a_1 < a_2 < \dots$ be the lengths of cycles in G . Is it true that $\sum \frac{1}{a_i} \gg \log k$? Is the sum $\sum \frac{1}{a_i}$ minimised when G is a complete bipartite graph?

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Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>